Stage 3 — Check-your-understanding Answers

Q1. The predictor–corrector weights by hand

Equations (19)–(20): b(k)=(k+1)^alpha-k^alpha; a(0,n)=n^(alpha+1)-(n-alpha)(n+1)^alpha; c(k)=(k+2)^(alpha+1)+k^(alpha+1)-2(k+1)^(alpha+1).

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alpha=0.5: b0=1^0.5-0^0.5=1

b1=sqrt(2)-1=0.414214

b2=sqrt(3)-sqrt(2)=0.317837

a(0,0)=0^1.5-(0-0.5)*1^0.5=0.5

c0=2^1.5+0^1.5-2*1^1.5=2sqrt(2)-2=0.828427
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At alpha=1: c(k)=(k+2)^2+k^2-2(k+1)^2=2 for every k. The corrector factor h^alpha/Gamma(alpha+2)=h/Gamma(3)=h/2. Also a(0,n)=1 and b(k)=1, giving the ordinary trapezoidal predictor–corrector scheme. For 0<alpha<1, b(k) decreases because k^alpha is concave; older history counts less but never exactly zero.

Q2. Why small alpha is stiffer

The conservative stability rule is h*lambda^(1/alpha) <= 1, so h_max=lambda^(-1/alpha). For the stipulated combined-rate bound lambda=1^alpha+1^alpha=2:

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alpha=1: h_max=2^(-1)=0.500000 weeks

alpha=0.5: h_max=2^(-2)=0.250000 weeks

alpha=0.3: h_max=2^(-10/3)=0.099213 weeks
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For a single removal rate gamma=2, lambda=2^alpha and h_max=(2^alpha)^(-1/alpha)=1/2 week for alpha=1, 0.5, and 0.3. The power cancels for a single rate, but (a^alpha+b^alpha)^(1/alpha) is not generally a+b. Smaller alpha can therefore magnify combined rates and require shorter steps.

Technical qualification: for the no-transmission (E,I) block with sigma=gamma=1, the actual eigenvalues are -1 and -1. The sum 2 used in the question is a conservative rate bound, not an eigenvalue of that block.

Q3. When L1 goes negative

At alpha=1, backward Euler for growth y' = lambda*y gives y1=y0/(1-h*lambda). Taking y0=1 and lambda=2 per week:

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h=0.25: y1=1/(1-0.25*2)=2

h=1: y1=1/(1-1*2)=-1
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For alpha=0.5, the L1 positivity condition is h^(-0.5)/Gamma(1.5)>2. Since Gamma(1.5)=sqrt(pi)/2:

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2/(sqrt(pi)*sqrt(h)) > 2

1/sqrt(h) > sqrt(pi) => h < 1/pi

h < 0.318310 weeks = 2.22817 days
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The boundary is excluded by the strict inequality. A negative numerical compartment is biologically impossible and can corrupt predicted incidence and fitted parameters; weekly data may therefore require sub-weekly internal

integration.

Q4. Reading the convergence table

The supplied manuscript labels these numbers Table 4 (not Table 5). At alpha=0.5, the PECE error falls from 7.9 x 10^-4 at h=0.1 to 9.3 x 10^-6 at h=0.00625. The step ratio is 16=2^4, so there are four halvings.

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error ratio = 0.00079/0.0000093 = 84.946237
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p = log2(84.946237)/4 = 1.602 (approx.)
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The smooth-theory benchmark is 1+alpha=1.5, so the endpoint-average order is about 0.102 higher. The paper reports 1.55 from the two smallest steps. These are observed fixed-time orders (at t=2), not a proof for the fractional test problem: its solution has t^alpha behavior near t=0, with limited smoothness, so smooth-function error theory does not apply directly.

Q5. What the decline can and cannot tell us

Figure 8 shows that the Caputo SEIR local log–log slope is strongly time-dependent. At alpha=0.8 it falls to about -4 shortly after the peak, rises to about -0.3 around week 450, is near -0.9 at week 1,000, and approaches -0.8 only much later. Both fractional models are near -alpha by about week 6,000 (slopes -0.612 at alpha=0.6 and -0.794 at alpha=0.8). These are numerical observations, not a nonlinear tail theorem.

Table 2 gives 33 mpox weeks, with the peak at week 12, leaving only about 20 observed weeks after the peak (weeks 13–32). A straight line fitted to this short log–log decline would measure a transient, window-dependent slope, not alpha. The data do not reach the asymptotic regime, and weekly reported cases also differ from the latent infectious population I(t) plotted in Figure 8.

What may remain informative is the overall late-tail shape: fractional models can exhibit persistent algebraic decay, whereas the paper finds exponential decay for Erlang and declining-transmission models once transmission is below replacement. Distinguishing those shapes requires a sufficiently long, well-observed decline and explicit comparison of fitted models; it cannot be decided from a single short-window slope. This motivates the Stage 6 model comparison.